Domácí úkol 1.1

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1. $Re\left(\langle \vec{u}, \vec{v} \rangle\right) = \frac{1}{4}\left(\|\vec{u}+\vec{v}\|^2 - \|\vec{u}-\vec{v}\|^2\right)$

$$\frac{1}{4}\left(\|\vec{u}+\vec{v}\|^2 - \|\vec{u}-\vec{v}\|^2\right) \stackrel{(*)}{=}$$

$(*)$: Skalární součin sama sebe je nezáporné realné číslo, čili platí $(\sqrt{<>})^2 = <>$

$$\stackrel{(*)}{=} \frac{1}{4}\left(\langle \vec{u}+\vec{v}, \vec{u}+\vec{v}\rangle - \langle \vec{u}-\vec{v}, \vec{u}-\vec{v}\rangle\right)$$
$$= \frac{1}{4}\Big(\left(\langle \vec{u},\vec{u}\rangle + \langle \vec{u},\vec{v}\rangle + \langle \vec{v},\vec{u}\rangle + \langle \vec{v},\vec{v}\rangle\right) - \left(\langle \vec{u},\vec{u}\rangle - \langle \vec{u},\vec{v}\rangle - \langle \vec{v},\vec{u}\rangle + \langle \vec{v},\vec{v}\rangle\right) \Big)$$
$$= \frac{1}{4}\left(2\langle \vec{u},\vec{v}\rangle + 2\langle \vec{v},\vec{u}\rangle\right) = \frac{1}{2}\left(\langle \vec{u},\vec{v}\rangle + \langle \vec{v},\vec{u}\rangle\right) = \frac{1}{2}\left(\langle \vec{u},\vec{v}\rangle + \overline{\langle \vec{u},\vec{v}\rangle}\right)$$
$$= \frac{1}{2}\left(Re\langle \vec{u},\vec{v}\rangle + i\cdot Im\langle \vec{u},\vec{v}\rangle + Re\langle \vec{u},\vec{v}\rangle - i\cdot Im\langle \vec{u},\vec{v}\rangle\right)$$
$$= \frac{1}{2}\left(2Re\langle \vec{u},\vec{v}\rangle\right) = \underline{\underline{Re\langle \vec{u},\vec{v}\rangle}} \qquad \square$$

2. $Im\left(\langle \vec{u}, \vec{v} \rangle\right) = \frac{1}{4}\left(\|\vec{u}-i\vec{v}\|^2 - \|\vec{u}+i\vec{v}\|^2\right)$

$$\frac{1}{4}\left(\|\vec{u}-i\vec{v}\|^2 - \|\vec{u}+i\vec{v}\|^2\right) \stackrel{(*)}{=} \frac{1}{4}\left(\langle \vec{u}-i\vec{v}, \vec{u}-i\vec{v}\rangle - \langle \vec{u}+i\vec{v}, \vec{u}+i\vec{v}\rangle\right)$$
$$= \frac{1}{4}\Big(\left(\langle \vec{u},\vec{u}\rangle - \langle \vec{u},i\vec{v}\rangle - \langle i\vec{v},\vec{u}\rangle + \langle i\vec{v},i\vec{v}\rangle\right) - \left(\langle \vec{u},\vec{u}\rangle + \langle \vec{u},i\vec{v}\rangle + \langle i\vec{v},\vec{u}\rangle + \langle i\vec{v},i\vec{v}\rangle\right) \Big)$$
$$= \frac{1}{4}\left(-2\langle \vec{u},i\vec{v}\rangle - 2\langle i\vec{v},\vec{u}\rangle\right) = \frac{-1}{2}\left(\langle \vec{u},i\vec{v}\rangle + \langle i\vec{v},\vec{u}\rangle\right) = \frac{-1}{2}\left(\langle \vec{u},i\vec{v}\rangle + \overline{\langle \vec{u},i\vec{v}\rangle}\right)$$
$$= \frac{-1}{2}\left(i\langle \vec{u},\vec{v}\rangle + \overline{i\langle \vec{u},\vec{v}\rangle}\right) = \frac{-i}{2}\left(\langle \vec{u},\vec{v}\rangle - \overline{\langle \vec{u},\vec{v}\rangle}\right)$$
$$= \frac{-i}{2}\left(Re\langle \vec{u},\vec{v}\rangle + i\cdot Im\langle \vec{u},\vec{v}\rangle - \left(Re\langle \vec{u},\vec{v}\rangle - i\cdot Im\langle \vec{u},\vec{v}\rangle\right)\right)$$
$$= \frac{-i}{2}\left(2i\cdot Im\langle \vec{u},\vec{v}\rangle\right) = \underline{\underline{Im\langle \vec{u},\vec{v}\rangle}} \qquad \square$$